

Implement new strain rate tensor

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→ I think we only need to update the scalar equations for $R_{\theta\theta}/R_{\phi\phi}$ interaction with the nonlinear strain rate.

→ Equations are still solved in the reference state.

→ We need to update, in particular, the membrane eq.

Observations:

→ Cotangent is singular at $\{0, \pi\}$.

→ Nonlinear strain rate already in axisymmetric coords.

New constitutive equation:

$$\begin{cases} \text{Before: } \sigma(u; K) = K \nabla_r(u) \\ \text{Now: } \sigma(u; K, \xi) = S(K, \xi) P + \\ \Lambda(K, \xi) \text{tr}(\epsilon(u)) P + \\ 2M(K, \xi) \epsilon(u) \end{cases}$$

→ Não se com passar a 1D?

- Axisymmetric w/ cylindrical coordinates

$$\begin{cases} \tilde{\epsilon}(u) = \begin{bmatrix} \frac{\partial u^r}{\partial r} & 0 & \frac{1}{2} \left(\frac{\partial u^r}{\partial z} + \frac{\partial u^z}{\partial r} \right) \\ 0 & \frac{u^r}{r} & 0 \\ \frac{1}{2} \left(\frac{\partial u^r}{\partial z} + \frac{\partial u^z}{\partial r} \right) & 0 & \frac{\partial u^z}{\partial z} \end{bmatrix} \\ \tilde{\epsilon}(u) \cdot \tilde{\epsilon}(u) = \dots \end{cases}$$

→ $R^2 \sin \theta$

- Tangent w/ spherical coordinates **UPDATE MEASURE !!!**

$$\left[\frac{\partial u^r}{\partial r} \quad \frac{1}{r} \frac{\partial u^r}{\partial \theta} \quad \frac{u^\theta}{r} \quad \frac{1}{r} \frac{\partial u^r}{\partial \phi} \quad \frac{u^\phi}{r} \right]$$

$$\tilde{\nabla}(\vec{u}) = \begin{pmatrix} \frac{\partial u^r}{\partial r} & \frac{1}{R} \frac{\partial u^r}{\partial \theta} - \frac{u^\theta}{R} & \frac{1}{R \sin \theta} \frac{\partial u^r}{\partial \varphi} - \frac{u^\varphi}{R} \\ \frac{\partial u^\theta}{\partial r} & \frac{1}{R} \frac{\partial u^\theta}{\partial \theta} - \frac{u^r}{R} & \frac{1}{R \sin \theta} \frac{\partial u^\theta}{\partial \varphi} - \cot \theta \frac{u^\varphi}{R} \\ \frac{\partial u^\varphi}{\partial r} & \frac{1}{R} \frac{\partial u^\varphi}{\partial \theta} & \frac{1}{R \sin \theta} \frac{\partial u^\varphi}{\partial \varphi} + \cot \theta \frac{u^\theta}{R} + \frac{u^r}{R} \end{pmatrix}$$

Assuming $\vec{u}(r, \theta, \varphi) = u^\theta(\theta) \hat{e}_\theta$

$$\tilde{\nabla}(\vec{u}) = \frac{1}{R} \begin{pmatrix} 0 & -u^\theta & 0 \\ 0 & \partial u^\theta / \partial \theta & 0 \\ 0 & 0 & \cot \theta u_\theta \end{pmatrix}$$

$$\tilde{\mathcal{E}}(\vec{u}) = \frac{1}{R} \begin{pmatrix} 0 & -u^\theta/2 & 0 \\ -u^\theta/2 & \partial u^\theta / \partial \theta & 0 \\ 0 & 0 & \cot \theta u_\theta \end{pmatrix}$$

$$\begin{aligned} \frac{1}{2} \tilde{\nabla}(\vec{u})^T \cdot \tilde{\nabla}(\vec{u}) &= \frac{1}{2R^2} \begin{pmatrix} 0 & 0 & 0 \\ -u^\theta & \partial u^\theta / \partial \theta & 0 \\ 0 & 0 & \cot \theta u_\theta \end{pmatrix} \begin{pmatrix} 0 & -u^\theta & 0 \\ 0 & \partial u^\theta / \partial \theta & 0 \\ 0 & 0 & \cot \theta u_\theta \end{pmatrix} \\ &= \frac{1}{2R^2} \begin{pmatrix} 0 & 0 & 0 \\ 0 & u^{\theta 2} + \partial u^\theta / \partial \theta^2 & 0 \\ 0 & 0 & (\cot \theta u_\theta)^2 \end{pmatrix} \end{aligned}$$

dropping θ in u^θ

$$\tilde{\mathcal{E}}(\vec{u}) : \tilde{\mathcal{E}}(\vec{u}) \stackrel{\downarrow}{=} \frac{1}{2} \int \frac{u_\theta^2}{r^2} + \frac{\partial u}{\partial r} \frac{\partial u}{\partial r} + (\cot \theta)^2 u_\theta^2 \quad \checkmark$$

$$\underline{\underline{\tilde{\mathcal{E}}(\vec{u}) : \tilde{\mathcal{E}}(\vec{v})}} \stackrel{(1)}{=} \frac{1}{R^2} \left\{ \frac{uv}{2} + \underbrace{\frac{\partial u}{\partial \theta} \frac{\partial v}{\partial \theta}}_{\text{surface gradient} = \nabla^d(u, n_T)} + (\cot \theta)^2 uv \right\} \checkmark$$

$$\underline{\underline{\frac{1}{2} (\vec{\nabla}(\vec{u})^T \cdot \vec{\nabla}(\vec{u})) : \tilde{\mathcal{E}}(\vec{v})}} = \int \underline{\underline{\text{div}(\mathcal{E}(\vec{u})) \cdot \vec{v}}} = \dots = \int \underbrace{\mathcal{E}(\vec{u}) : \tilde{\mathcal{E}}(\vec{v})}_{(1)+(2)}$$

$$\frac{1}{2R^3} \begin{pmatrix} 0 & 0 & 0 \\ 0 & u^2 + \left(\frac{\partial u}{\partial \theta}\right)^2 & 0 \\ 0 & 0 & (\cot \theta u)^2 \end{pmatrix} \begin{pmatrix} 0 & -v/2 & 0 \\ -v/2 & \partial v / \partial \theta & 0 \\ 0 & 0 & \cot \theta v \end{pmatrix} =$$

$$\frac{1}{2R^3} \left\{ \left[u^2 + \left(\frac{\partial u}{\partial \theta}\right)^2 \right] \frac{\partial v}{\partial \theta} + (\cot \theta u)^2 \cot \theta v \right\} \checkmark$$

$\hookrightarrow$ to linearise

(3) for t_n , evaluate at t_{n-1}

$$\text{tr } \tilde{\mathcal{E}}(\vec{u}) = \frac{1}{R} \left(\frac{\partial u}{\partial \theta} + \cot \theta u \right)$$

$$\text{Id} : \mathcal{E}(\vec{v}) = \underline{\underline{\text{tr } \tilde{\mathcal{E}}(\vec{v})}} = \frac{1}{R} \left(\frac{\partial v}{\partial \theta} + \cot \theta v \right)$$

$$\left(\text{tr } \tilde{\mathcal{E}}(\vec{u}) \underline{\underline{\text{Id} : \mathcal{E}(\vec{v})}} \right) =$$

$$\frac{1}{R^2} \left\{ \frac{\partial u}{\partial \theta} \frac{\partial v}{\partial \theta} + \cot \theta u \frac{\partial v}{\partial \theta} + \frac{\partial u}{\partial \theta} \cot \theta v + (\cot \theta)^2 uv \right\}$$

$$\text{tr} \left(\frac{1}{2} \vec{\nabla}(\vec{u})^T \cdot \vec{\nabla}(\vec{u}) \right) = \frac{1}{2R^2} \left(\underbrace{[1 + (\cot \theta)^2]}_{(3)} \underbrace{u^2}_{(3)} + \underbrace{\left[\frac{\partial u}{\partial \theta} \right]^2}_{(3)} \right)$$

$$\text{tr} \left(\frac{1}{2} \vec{\nabla}(\vec{u})^T \cdot \vec{\nabla}(\vec{u}) \right) \text{Id} : \mathcal{E}(\vec{v}) =$$

$$(2 \dots \dots)$$

$$\frac{1}{2R^3} \left\{ \left(\underbrace{[1 + (\cot \theta)^2] u^2}_{(3)} + \underbrace{\left[\frac{\partial u}{\partial \theta} \right]^2}_{(3)} \right) \left(\frac{\partial \varphi}{\partial \theta} + \cot \theta \varphi \right) \right\}$$

$$\int \text{tr}(E(\vec{x})) \mathbb{I}_A : E(\vec{x})$$